

Experiment 6 — Managed ML Services using Amazon SageMaker and Rekognition

Part A: Amazon SageMaker Exploration

Step 1: Access SageMaker

1. Log in to **AWS Management Console**
2. In the top search bar, type **SageMaker**
3. Click on **Amazon SageMaker**
4. The SageMaker Dashboard will open

 **Screenshot needed:** SageMaker dashboard showing the overview panel with Environment, Applications, Model Training sections

Step 2: Understand SageMaker Components

In the left navigation pane, explore and note the following sections:

Component	Purpose
SageMaker Studio	Fully integrated IDE for ML — create notebooks, run experiments, manage deployments
Notebooks	Write and execute Python code for ML workflows using Jupyter
Training Jobs	Train ML models on managed compute resources without manual server setup
Endpoints	Deploy trained models for real-time prediction and inference
Built-in Algorithms / JumpStart	Pre-built algorithms to train models quickly without writing code from scratch

 **Screenshot needed:** Left navigation pane of SageMaker showing these sections

Step 3: Create SageMaker Domain and Launch Studio

1. On the SageMaker dashboard, click **Domains** in the left pane under Environment Configuration

2. Click **Create Domain**
3. Select **Quick Setup** (default settings)
4. Enter required details if prompted (domain name, user profile name)
5. Click **Set up** and wait for domain creation to complete (may take a few minutes)
6. Once created, click **Launch** next to your domain
7. Select **Studio**
8. Wait for SageMaker Studio to open in a new browser tab

 Screenshots needed:

- SageMaker Studio landing page showing "The first fully integrated development environment for machine learning"
 - SageMaker Studio environment open showing the Overview/Notebooks panel
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Step 4: Explore Notebook Section

1. Inside SageMaker Studio, click **Notebooks** in the left sidebar
2. Click **Create notebook instance** to observe the options available
3. Note that a notebook instance named **myNotebook** with instance type **ml.t3.medium** can be created here
4. Observe the interface — notebooks support Python, SQL, Markdown, and Charts

 **Screenshot needed:** Notebook Instances page showing either an existing notebook or the Create notebook instance screen

Step 5: Explore Training Jobs and Endpoints

1. In the left navigation pane, click **Training Jobs** (under Model Training & Customization)
2. Observe the training jobs list — it will show "No training jobs found" if none have been run
3. Note that training jobs are used to train ML models on managed compute resources
4. In the left navigation pane, click **Endpoints** (under Deployments & Inference)
5. Observe the endpoints list — it will show "There are currently no resources"
6. Note that endpoints are used to deploy trained models for real-time prediction

 Screenshots needed:

- Training Jobs page
 - Endpoints page
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Part B: Amazon Rekognition Demo

Step 6: Access Amazon Rekognition

1. Open a new tab and go to **AWS Management Console**
2. In the search bar, type **Rekognition**
3. Click on **Amazon Rekognition**
4. The Rekognition dashboard will open showing available demo options in the left panel



Screenshot needed: Amazon Rekognition dashboard/landing page

Step 7: Perform Label Detection

1. In the left sidebar under **Demos**, click **Label Detection**
2. The Label Detection demo page will open
3. You will see sample images provided by AWS — click any one of them, OR
4. Click **Upload** or **drag and drop** your own image (any photo — a car, animal, person, scenery etc.)
5. Wait for Rekognition to analyze the image
6. Observe the **Results** panel on the right showing:
 - Detected labels (e.g., Car, Vehicle, Transportation)
 - Confidence scores for each label (e.g., 99.8%, 98.7%)



Screenshots needed:

- Label Detection page with your uploaded image and results showing detected labels and confidence scores
 - Try with two different images if possible (e.g., an anime/cartoon image and a real photo to show varied detection results)
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Step 8: Record Observations

Note the following in your journal:

Observation	Detail
Model Training Required	No — Rekognition uses pre-trained deep learning models
Service Type	API-driven ML service
Result Speed	Immediate — results generated instantly after image upload

Scalability	Automatic — scales with request volume without user intervention
Detected Labels	Note the actual labels and confidence scores from your image

Final Summary:

Search Sagemaker AI and create Domain
Open Notebooks and create a notebook instance
Show training jobs and endpoints from sidebar
Search Rekognition and perform label detection between two images